

Cloud Computing in Computer Networking

What is Cloud Computing?

Cloud Computing is the delivery of computing services such as servers, storage, databases, networking, software, analytics, and intelligence over the Internet ("the cloud") instead of using local computers or on-premises data centers.

Instead of purchasing and maintaining physical hardware, users can access resources on demand from cloud providers and pay only for what they use.

Definition

Cloud Computing is a model that enables convenient, on-demand network access to a shared pool of configurable computing resources that can be rapidly provisioned and released with minimal management effort.

Why Cloud Computing?

Traditional IT Infrastructure requires:

- Buying physical servers
- Setting up networking equipment
- Maintaining hardware
- Managing storage
- Handling backups and security

Cloud Computing solves these problems by providing resources over the internet.

Traditional Model

User



Company Data Center



Physical Servers



Applications

Cloud Model

User



Internet



Cloud Provider



Virtual Resources

Characteristics of Cloud Computing

1. On-Demand Self-Service

Users can provision resources whenever needed without human intervention.

Example:

- Create a virtual machine in minutes.

2. Broad Network Access

Services are available over the internet and accessible from:

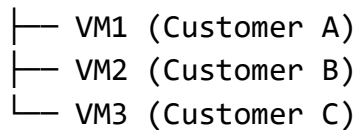
- Computers
- Laptops
- Tablets
- Smartphones

3. Resource Pooling

Cloud providers share resources among multiple customers.

Example:

Physical Server



4. Rapid Elasticity

Resources can scale up or down automatically.

Example:

Normal Traffic



2 Servers

High Traffic



20 Servers

5. Measured Service

Users pay only for consumed resources.

Examples:

- Storage used
- CPU hours
- Network bandwidth

How Cloud Computing Works

Cloud computing relies heavily on networking and virtualization.

Step 1: User Request

User accesses a cloud application.

User → Internet → Cloud Service

Step 2: Request Reaches Data Center

The request travels through routers, switches, and load balancers to the cloud provider's data center.

Step 3: Virtualization Layer

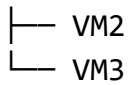
A hypervisor creates multiple Virtual Machines (VMs) on physical hardware.

Physical Server



Hypervisor

├─ VM1



Examples:

- VMware ESXi
- Hyper-V
- KVM

Step 4: Resource Allocation

Cloud management software allocates:

- CPU
- RAM
- Storage
- Network bandwidth

based on demand.

Step 5: Service Delivery

The requested application or service is delivered back to the user via the internet.

Role of Networking in Cloud Computing

Networking is the backbone of cloud computing.

Networking Components

Routers

Forward packets between networks.

Switches

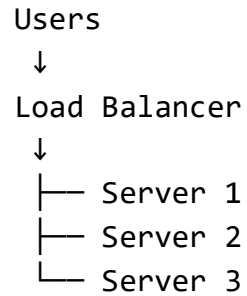
Connect devices within a network.

Firewalls

Protect cloud resources from attacks.

Load Balancers

Distribute traffic among servers.



Virtualization

Virtualization allows multiple virtual machines to run on a single physical server.

Without Virtualization

Server 1 → App A

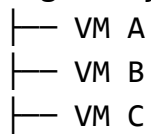
Server 2 → App B

Server 3 → App C

Many resources remain unused.

With Virtualization

Single Physical Server



Better resource utilization.

Service Models of Cloud Computing

1. Infrastructure as a Service (IaaS)

Provides virtualized computing infrastructure.

Users manage:

- Operating System
- Applications
- Data

Provider manages:

- Hardware
- Networking
- Virtualization

Examples:

- Amazon Web Services EC2
- Microsoft Virtual Machines
- Google Compute Engine

Architecture

Applications

Operating System

Virtual Machines

Cloud Infrastructure

2. Platform as a Service (PaaS)

Provides a platform for application development.

Users manage:

- Code
- Application Logic

Provider manages:

- Servers
- OS
- Runtime
- Networking

Examples:

- Google App Engine
- Microsoft App Service

Architecture

Application Code

Runtime

OS

Infrastructure

3. Software as a Service (SaaS)

Ready-to-use software delivered via internet.

Users simply use the application.

Examples:

- [Gmail](#)
- [Google Docs](#)
- [Microsoft 365](#)

Architecture

User



Browser



Cloud Application

Deployment Models

1. Public Cloud

Resources are owned by cloud providers and shared among customers.

Examples:

- [AWS](#)
- [Azure](#)
- [Google Cloud](#)

Advantages

- Low cost
- Easy scalability
- No maintenance

2. Private Cloud

Dedicated cloud environment for one organization.

Organization



Private Data Center

Advantages

- Better control
- Higher security
- Custom configurations

3. Hybrid Cloud

Combination of public and private clouds.

Private Cloud



Public Cloud

Used when sensitive data remains private while public cloud handles extra workload.

4. Community Cloud

Shared among organizations with common requirements.

Example:

- Universities
- Government agencies

Benefits of Cloud Computing

Cost Efficiency

No need to purchase expensive hardware.

Scalability

Resources can increase or decrease instantly.

High Availability

Cloud providers use multiple data centers.

Data Center A



Failover



Data Center B

Disaster Recovery

Data backups are maintained automatically.

Global Access

Services can be accessed from anywhere with internet connectivity.

Faster Deployment

Applications can be deployed in minutes.

Challenges of Cloud Computing

Security Risks

- Data breaches
- Unauthorized access
- Insider threats

Internet Dependency

No internet means limited cloud access.

Vendor Lock-in

Migrating from one cloud provider to another may be difficult.

Compliance Issues

Organizations must follow regulations regarding data storage and privacy.

Security in Cloud Computing

Encryption

Protects data during storage and transmission.

Identity and Access Management (IAM)

Controls who can access cloud resources.

Firewalls

Protect cloud networks from attacks.

Multi-Factor Authentication (MFA)

Provides additional security beyond passwords.

Monitoring and Logging

Tracks suspicious activities and system events.

Real-World Example

Imagine a company hosting an e-commerce website:

Customer



Internet



Load Balancer



Web Servers



Application Servers



Database

During a sale:

- Traffic increases dramatically.
- Cloud automatically creates additional servers.
- Load balancer distributes traffic.
- When traffic decreases, extra servers are removed.

This is called **Auto Scaling**, one of the biggest advantages of cloud computing.

Advantages vs Disadvantages

Advantages

Scalability
Cost-effective
High availability
Easy deployment
Automatic backup

Disadvantages

Internet dependency
Vendor lock-in
Security concerns
Compliance challenges
Less direct control